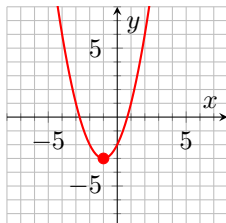


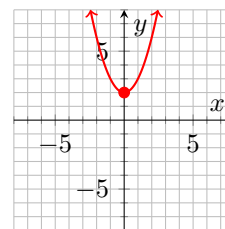
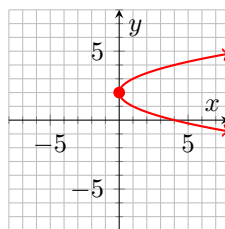
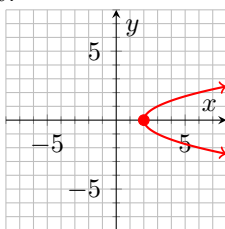
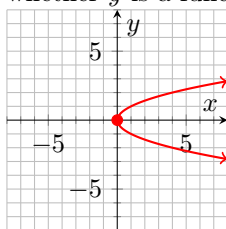
Determine the domain and range of each relation, and state whether the relation is a function.

1. $f = \{(10, 1), (27, -1), (37, 1), (10, 2), (47, 2)\}$

2. The curve shown in the graph below.



3. $x = y^2 + 2$. Choose the correct graph of the equation, determine the domain and range, and determine whether y is a function of x .



Let $g(x) = -x^2 + 5x + 3$.

4. Find $g(-2)$

Use the difference quotient, $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$, to complete and simplify the following problem.

5. $f(x) = x^2 + 1$.